

2. human MARCKS REGION "Calmodulin-binding (PSD)" (25 aa) and zebrafish Q6PD99

TOOLS: kmerseek hp_thomas_dill2 k19 (chosen for REGION in Stage 0) and hp_lehninger2 k18; phmmer, MMseqs2, MMseqs2 iterative, Foldseek, ProstT5, Reseek; Kyte-Doolittle scan

- kmerseek: call lands on the feature (>= 80% of call inside, >= 30% covered)

call mostly inside the feature, lower IoU than kmerseek

call overlaps the feature but < 50% of the call is inside it

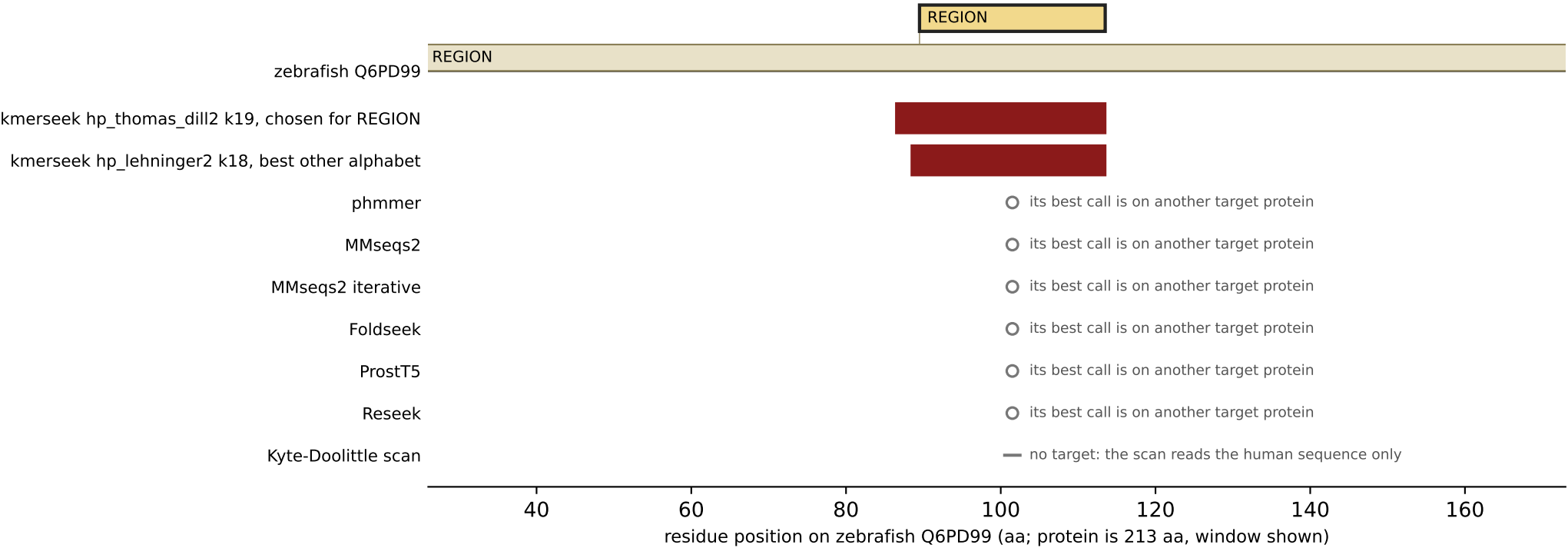
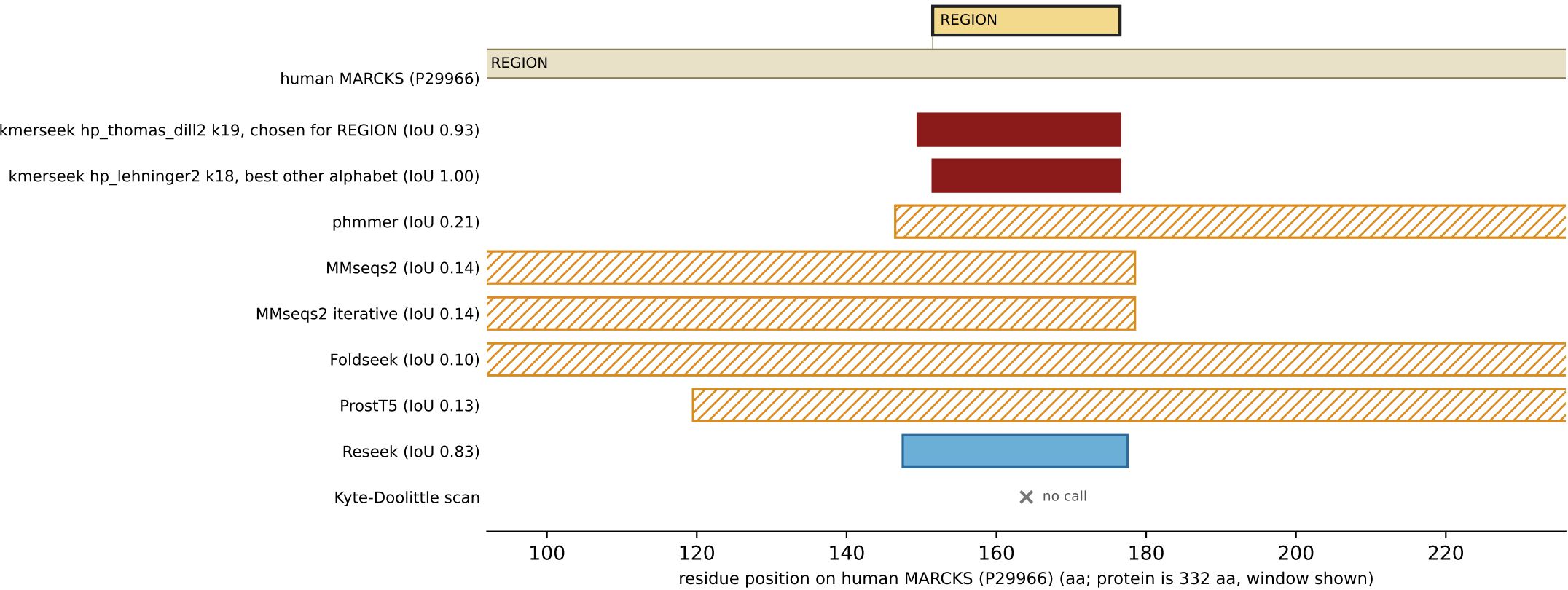
call reaches kmerseek's IoU or higher

no call of this feature type overlaps the feature
- the tool's best call is on another target protein

Kyte-Doolittle scan: no target protein, it searches nothing

the Swiss-Prot feature being scored (human) / transferred from (target)

other Swiss-Prot features, labelled by type



Hypothesis: kmerseek places the human REGION feature "Calmodulin-binding (PSD)" on its zebrafish counterpart and the structure tools do not.
Conclusion: kmerseek IoU 0.93, 17 of 27 residues identical; Foldseek spills; ProstT5 spills; Reseek inside, lower IoU; phmmer spills; MMseqs2 spills; MMseqs2 iterative spills; Kyte-Doolittle scan did not land; pLDDT 70, disorder fraction 1.0.